

CHAPTER II.6, WEEK 8

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6.1. Let X be a scheme satisfying $(*)$. Then $X \times \mathbb{P}^n$ also satisfies $(*)$, and $\text{Cl}(X \times \mathbb{P}^n) \cong \text{Cl}(X) \times \mathbb{Z}$.

Proof. We note that $X \times \mathbb{P}^n$ is covered by finitely many $X \times U_i \cong X \times \mathbb{A}^n$. $X \times \mathbb{A}^n$ satisfies $(*)$ by (6.6) and induction on n . It follows that $X \times \mathbb{P}^n$ is noetherian, reduced, and regular in codimension one. Each $(X \times U_i) \cap (X \times U_j) = X \times (U_i \cap U_j)$ is nonempty, so $X \times \mathbb{P}^n$ is irreducible, and hence integral. It suffices, then, to show that $X \times \mathbb{P}^n$ is separated.

Let $\text{pr}_2 : X \times \mathbb{P}^n \rightarrow \mathbb{P}^n$ denote the projection. Note that our cover $\{U_i\}$ has $\text{pr}_2^{-1}(U_i) = X \times U_i$. The maps $X \times U_i \rightarrow U_i \rightarrow \text{Spec } \mathbb{Z}$ are separated, so that $X \times U_i \rightarrow U_i$ is separated. It follows that $X \times \mathbb{P}^n \rightarrow \mathbb{P}^n$ is separated, so $\mathbb{P}^n \rightarrow \text{Spec } \mathbb{Z}$ separated gives us that $X \times \mathbb{P}^n$ is separated.

Consider the closed subset $H := X \times V(x_0) \subset X \times \mathbb{P}^n$. It is easily seen to be irreducible of codimension one, so that (6.5) gives the following exact sequence:

$$\mathbb{Z} \rightarrow \text{Cl}(X \times \mathbb{P}^n) \rightarrow \text{Cl}(X \times U_0) \rightarrow 0.$$

(6.6) and induction on n once again gives us $\text{Cl}(X \times U_0) \cong \text{Cl}(X)$. We begin by showing $\mathbb{Z} \rightarrow \text{Cl}(X \times \mathbb{P}^n)$ is injective. We will first show that $H \in \text{CaCl}(X \times \mathbb{P}^n) \cong \text{Pic}(X \times \mathbb{P}^n)$, then show $\mathcal{L}(H) \mapsto 1 \in \mathbb{Z}$ under a series of maps.

For any open affine $\text{Spec } A \cong V \subset X$, $V \times \mathbb{P}^n \cong \mathbb{P}_A^n$, and $H|_{V \times U_i} = (x_0)$ for all i, j with $j \neq 0$. It follows that $H \in \text{CaCl}(X \times \mathbb{P}^n) \cong \text{Pic}(X \times \mathbb{P}^n)$, and $\mathcal{L}(H)|_{V \times \mathbb{P}^n} \cong \mathcal{O}_{\mathbb{P}_A^n}(1)$. Now fix a particular $\text{Spec } A \cong V \subset X$, and a point $x \in V$. We can base extend $\text{Spec } k(x) \rightarrow \text{Spec } A$ to a map $\mathbb{P}_{k(x)}^n \rightarrow \mathbb{P}_A^n$. Composing with $\mathbb{P}_A^n \rightarrow X \times \mathbb{P}^n$ gives us a map $f : \mathbb{P}_{k(x)}^n \rightarrow X \times \mathbb{P}^n$.

(Ex. 6.8) gives us a map $f^* : \text{Pic}(X \times \mathbb{P}^n) \rightarrow \text{Pic}(\mathbb{P}_{k(x)}^n) \cong \mathbb{Z}$ such that

$$\mathcal{L}(H) \mapsto \mathcal{O}_{k(x)}^n \mapsto 1 \in \mathbb{Z}.$$

Then $m \cdot H \mapsto m \in \mathbb{Z}$ under this series of maps, so that in particular $m \cdot H \neq 0$. Hence, $\mathbb{Z} \rightarrow \text{Cl}(X \times \mathbb{P}^n)$ is injective. Note as well that the above defines a map $\text{Cl}(X \times \mathbb{P}^n) \rightarrow \text{Cl}(\mathbb{P}_{k(x)}^n) \cong \mathbb{Z}$ such that the following diagram, which we have just shown is exact, commutes.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \text{Cl}(X \times \mathbb{P}^n) & \longrightarrow & \text{Cl}(X) \longrightarrow 0 \\ & & & \swarrow \cong & \downarrow & & \\ & & & & \text{Cl}(\mathbb{P}_{k(x)}^n) & & \end{array}$$

The splitting lemma then implies that, in fact, $\text{Cl}(X \times \mathbb{P}^n) \cong \text{Cl}(X) \times \mathbb{Z}$. □

6.5. *Quadric Hypersurfaces.* Let $\text{char } k$, and let X be the affine quadric hypersurface $\text{Spec } k[x_0, \dots, x_n]/(x_0^2 + x_1^2 + \dots + x_r^2)$ —cf. (I, Ex. 5.12).

(a) Show that X is normal if $r \geq 2$ (use (Ex. 6.4)).

(b) Show by a suitable linear change of coordinates that the equation of X could be written as $x_0 x_1 = x_2^2 + \dots + x_r^2$. Now, imitate the method of (6.5.2) to show that:

- (1) If $r = 2$, then $\text{Cl } X \cong \mathbb{Z}/2\mathbb{Z}$;
- (2) If $r = 3$, then $\text{Cl } X \cong \mathbb{Z}$ (use (6.6.1) and (Ex. 6.3) above);
- (3) If $r \geq 4$, then $\text{Cl } X = 0$.

Proof. (a) Since $r \geq 1$, $f := x_1^2 + \cdots + x_r^2$ is non-constant. We will now show that f is square-free. Suppose $f = g^2$ for some $g \in k[x_1, \dots, x_r]$. f is of degree 2, so $g = a + \sum_i a_i x_i$ for some $a, a_i \in k$. Then $f = g^2$ implies $a^2 = 0$, and $2a_i a_j = 0$ for all $i \neq j$. $\text{char } k \neq 2$ implies $a_i a_j = 0$ for all $i \neq j$, so that at most one of the a_i is nonzero. Then $f = a_i^2 x_i^2$ implies f is a monomial, i.e. $r = 1$ and $f = x_1^2$. This contradicts $r \geq 2$, so f is indeed square-free.

Clearly $-f$ is non-constant and square-free as well. Thus, we may apply (Ex. 6.4) to show

$$\begin{aligned} A &= \frac{k[x_0, \dots, x_r]}{(x_0^2 + x_1^2 + \cdots + x_r^2)} \\ &= \frac{k[x_1, \dots, x_r][x_0]}{(x_0^2 - (-f))} \end{aligned}$$

is integrally closed, and hence normal. Then X is the spectrum of a normal ring, thus normal itself.

(b) The map given by $x_0 \mapsto \frac{i}{2}(y_0 + y_1)$, $x_1 \mapsto \frac{1}{2}(y_0 - y_1)$, and $x_i \mapsto y_i$ for all $1 < i \leq r$ is our desired change of coordinates. However, note that it is only well defined if $t^2 + 1$ splits in k , as x_0 maps to an element of $k(i)[y_0, y_1]$.

Clearly X is noetherian and separated. If $r \geq 2$, then (I,5.12b) gives us X integral and (a) gives us X normal, whence regular in codimension 1. If $r = 2$, then $X = \text{Spec } k[x, y, z]/(xy - z^2)$, so (1) is exactly (6.5.2).

Next, let $r = 3$. The same change of coordinates allows us to write $A = k[x, y, z, w]/(xy - zw)$, so that X is the cone of the surface Q defined in (6.6.1). Fix the hyperplane $H := V(x) \subset \mathbb{P}_k^3$, which does not contain Q . (Ex. 6.3b) gives the exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \text{Cl } Q \rightarrow \text{Cl } X \rightarrow 0,$$

where $1 \mapsto Q.H \in \text{Cl } Q$. $H \cap Q$ corresponds to the hyperplane $\{*\} \times \mathbb{P}^1$ under the isomorphism $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$, so $Q.H \mapsto (1, 0)$ under $\text{Cl } Q \cong \mathbb{Z} \oplus \mathbb{Z}$. Thus,

$$\text{Cl } X \cong \frac{\mathbb{Z} \oplus \mathbb{Z}}{(1, 0)\mathbb{Z}} \cong \mathbb{Z}$$

proves (3).

Finally, consider the case $r \geq 4$. Use our new coordinates to write $A = k[x_0, \dots, x_r]/(x_0^2 + \cdots + x_r^2 - x_0 x_1)$, and consider the closed subscheme

$$Y = \text{Spec } A/(x_0) \cong \text{Spec } k[x_1, x_2, \dots, x_r]/(x_2^2 + \cdots + x_r^2)$$

of X . $r - 2 \geq 2$, so $x_2^2 + \cdots + x_r^2$ is irreducible in $k[x_2, \dots, x_r]$. Then

$$A/(x_0) \cong (k[x_2, \dots, x_r]/(x_2^2 + \cdots + x_r^2))[x_1]$$

is an integral domain, so $x_2^2 + \cdots + x_r^2$ is irreducible in $k[x_1, \dots, x_r]$. Recalling that $A \cong k[x_0, \dots, x_r]/(x_0^2 + \cdots + x_r^2)$ is quotient by an irreducible ideal as well, Krull's Hauptidealsatz gives us that $\dim A = r$ and $\dim(A/(x_0)) = r - 1$. Since $A/(x_0)$ is integral, $(x_0) \trianglelefteq A$ is prime, so (1.8A) gives us $\text{codim}_X Y = \text{ht}(x_0) = 1$. Hence, Y is a prime divisor of X .

By the above, (6.5) supplies us with the exact sequence

$$\mathbb{Z} \rightarrow \text{Cl } X \rightarrow \text{Cl}(X - Y) \rightarrow 0,$$

where $1 \mapsto 1 \cdot Y$. Note that (x_0) is the generic point of Y , so the local ring of Y is given by $\mathcal{O}_{X,(x_0)}$. The uniformizer is $\mathcal{O}_{X,(x_0)}(x_0)$, so clearly $v_Y(x_0) = 1$. Clearly $v_{Y'}(x_0) = 0$ for all other prime divisors Y' , so $Y = (x_0)$ is principle. Thus, the exact sequence gives an isomorphism $\text{Cl } X \cong \text{Cl}(X - Y) = \text{Cl } D(x_0)$. Recall that $D(x_0) \cong \text{Spec } A_{x_0}$. We can write

$$\begin{aligned} A_{x_0} &= \frac{k[x_0, x_0^{-1}, x_1, x_2, \dots, x_r]}{x_2^2 + \dots + x_r^2 - x_0 x_1} \\ &\cong \frac{k[x_0, x_0^{-1}, x_1, x_2, \dots, x_r]}{x_0^{-1}(x_2^2 + \dots + x_r^2) - x_0} \\ &\cong k[x_0, x_0^{-1}, x_2, x_3, \dots, x_r]. \end{aligned}$$

This is clearly a UFD, so we have that

$$\text{Cl } X \cong \text{Cl } A_{x_0} = 0.$$

This proves (4). □

6.10 *The Grothendieck Group $K(X)$* . Let X be a noetherian scheme. We define $K(X)$ to be the quotient of the free abelian group generated by all coherent sheaves on X , by the subgroup generated by all expressions $\mathcal{F} - \mathcal{F}' - \mathcal{F}''$, whenever there is an exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ of coherent sheaves on X . If $\mathcal{F}$ is a coherent sheaf, we denote by $\gamma(\mathcal{F})$ its image in $K(X)$.

- (a) If $X = \mathbb{A}_k^1$, then $K(X) \cong \mathbb{Z}$.
- (b) If X is any integral scheme, and $\mathcal{F}$ a coherent sheaf, we define the rank of $\mathcal{F}$ to be $\dim_K \mathcal{F}_\xi$, where ξ is the generic point of X , and $K = k(\xi)$ is the function field of X . To show that the rank function defines a surjective homomorphism $\text{rk} : K(X) \rightarrow \mathbb{Z}$.
- (c) If Y is a closed subscheme of X , there is an exact sequence

$$K(Y) \rightarrow K(X) \rightarrow K(X - Y) \rightarrow 0,$$

where the first map is extension by zero, and the second map is restriction. [Hint: For exactness in the middle, show that if $\mathcal{F}$ is a coherent sheaf on X whose support is contained in Y , then there is a finite filtration $\mathcal{F} = \mathcal{F}_0 \supset \mathcal{F}_1 \supset \dots \supset \mathcal{F}_n = 0$, such that each $\mathcal{F}_i/\mathcal{F}_{i+1}$ is an $\mathcal{O}_Y$ -module. To show surjectivity on the right, use (Ex. 5.15).]

Proof. □